

Container Services



Is an App a good idea for my business?

Virtual Server 1



Virtual Server 2



Virtual Server 3



Virtualization Software (Example - VMware)



Physical Server

- VM – Issues
- Own OS, CPU, RAM and Storage
- Patching (Patch Tuesday)
- Monitoring
- License

Hello Containers

- Google (Long Time)
- All containers = One OS
- No license
- Less CPU, RAM and storage
- IBM System/360
- BSD Jails
- Solaris zones

Hardware layer
(Potentially a virtual machine)

Kernel space

User Space (Programs, etc.)

Cgroups (control group) = resource management

Namespaces = Process Isolation

Namespace

- Namespace isolates the resources of system, currently there are 6 kinds of namespaces in linux kernel.
 - Mount namespace
 - UTS namespace
 - IPC namespace
 - Net namespace
 - Pid namespace
 - User namespace

Namespaces

Development took over a decade: Namespaces implementation started in about **2002**; the last one, true for today, (user namespaces) was completed in February **2013**, in kernel 3.18.

There are currently 6 namespaces in Linux:

- **mnt** (mount points, filesystems)
- **pid** (processes)
- **net** (network stack)
- **ipc** (System V IPC)
- **uts** (hostname)
- **user** (UIDs)

In the past there were talks on adding more namespaces – device namespaces (LPC 2013), and other (OLS 2006, Eric W. Biederman).

**Hardware layer
(Potentially a virtual machine)**

Kernel space

**Network, PID, etc.
subsystems and their
namespaces**

Filesystems

**Namespace
1**

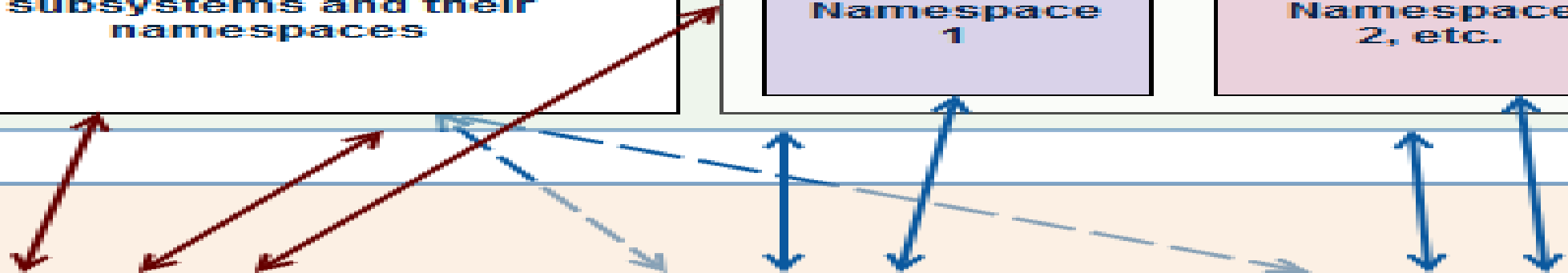
**Namespace
2, etc.**

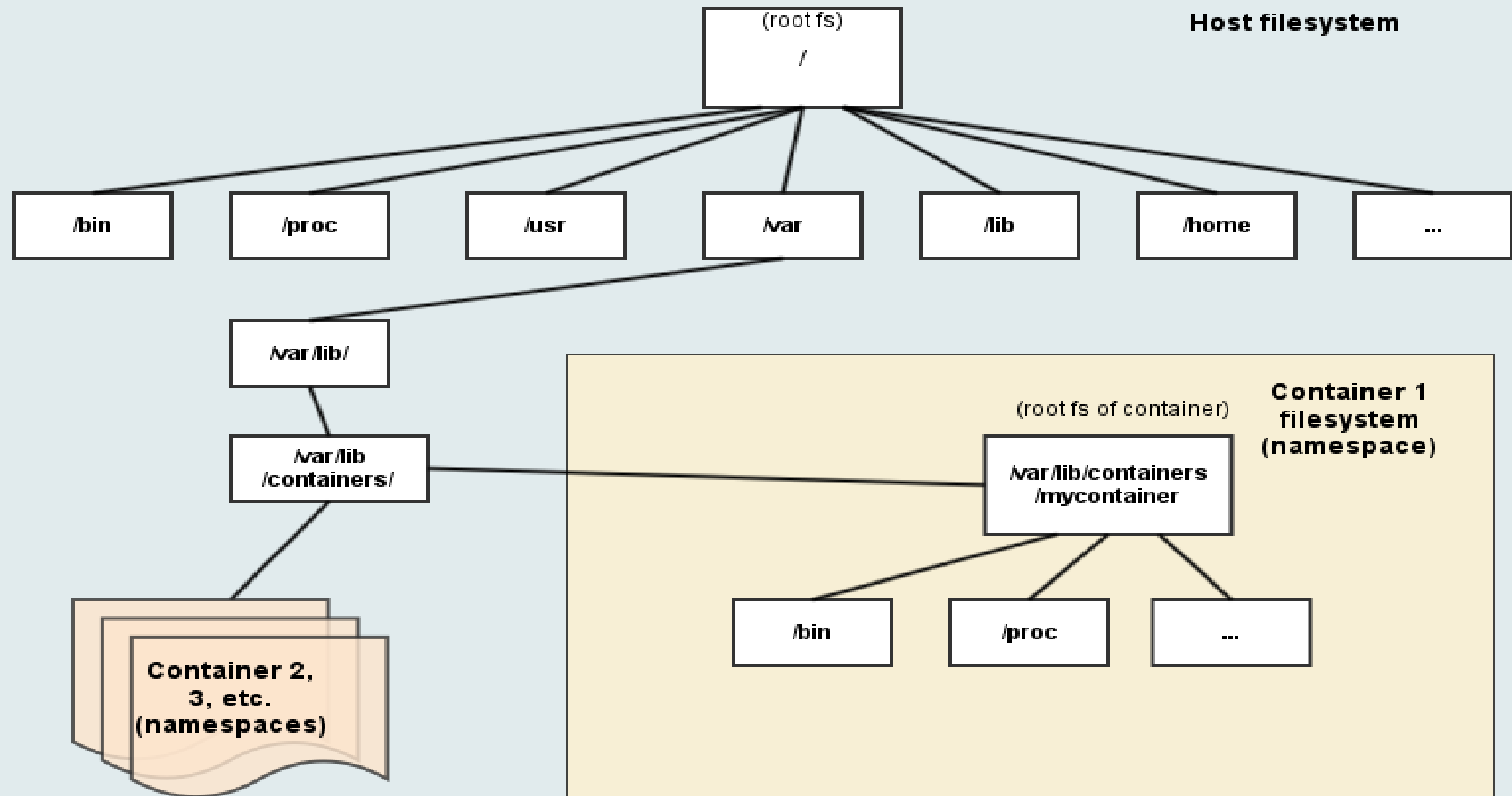
Host processes

Container 1

Container 2, etc.

User Space (Programs, etc.)





APP 1

BINS/LABS

APP 2

BINS/LABS

APP 3

BINS/LABS

DOCKER ENGINE

HOST OPERATING SYSTEM

INFRASTRUCTURE

Docker = Dock + Worker

- Docker Inc (Company) = 425 ++
- Docker (Container Runtime & Orchestration Technology)
- Docker (Open source Project) = Moby

Docker Inc (Mirantis)

- San Francisco
- Solomon Hykes (French-Born) American Developer
- Paas (dot cloud)
- Docker Conf (Docker con = 2019 = April 29 – May 2 (San Francisco))
- Docker Conf (Docker con = 2020 = Nov 17 – Nov 20)

Docker runtime & Orchestration Engine

- Community Edition (4 months = support)
- Enterprise Edition (12 months = support)
- Version = YY.MM-xx (18.06.0-ce)
- 18.06.1-ce
- Prior (2017) = major.minor (1.13) = last
- Quaterly releases
- Latest = 19.03.13

Docker Open-Source Project (Moby)

- Dockercon 2017 = Moby
- <https://github.com/moby>
- Golang (system-level programming) from Google

Batteries Included but removable

Docker client

docker

Docker Engine

Plug ins



Networking



Microsoft

vmware

CISCO

midokura



nuagenetworks



Volumes



ClusterHQ

Container Data Management

- CoreOS = appc (new open standard) and defined image format & container runtime
- Rkt (rocket)
- “Healthy competition is the mother of invention”

What is the Open Container Initiative (OCI)



An open source community (hosted by the Linux Foundation) for building a vendor-neutral, portable and open specification/runtime that delivers on the promise of containers as a source of application portability backed by a certification program.



OPEN CONTAINER
INITIATIVE



Container Ecosystem

Layer 5
CMP



DOCKER
CLOUD



Google Container Engine



DC/OS



OPENSIFT[®]
by Red Hat[®]

Layer 4
Orchestration



kubernetes



Docker Swarm



Tool Chain

flannel

clair



OPEN CONTAINER INITIATIVE

etcd

cAdvisor

Layer 3
Container
Engine



docker



LXD



rkt

Layer 2
Operating System

ubuntu[®]



redhat.
L I N U X



CoreOS

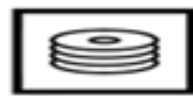
Layer 1
Infrastructure



Raw compute



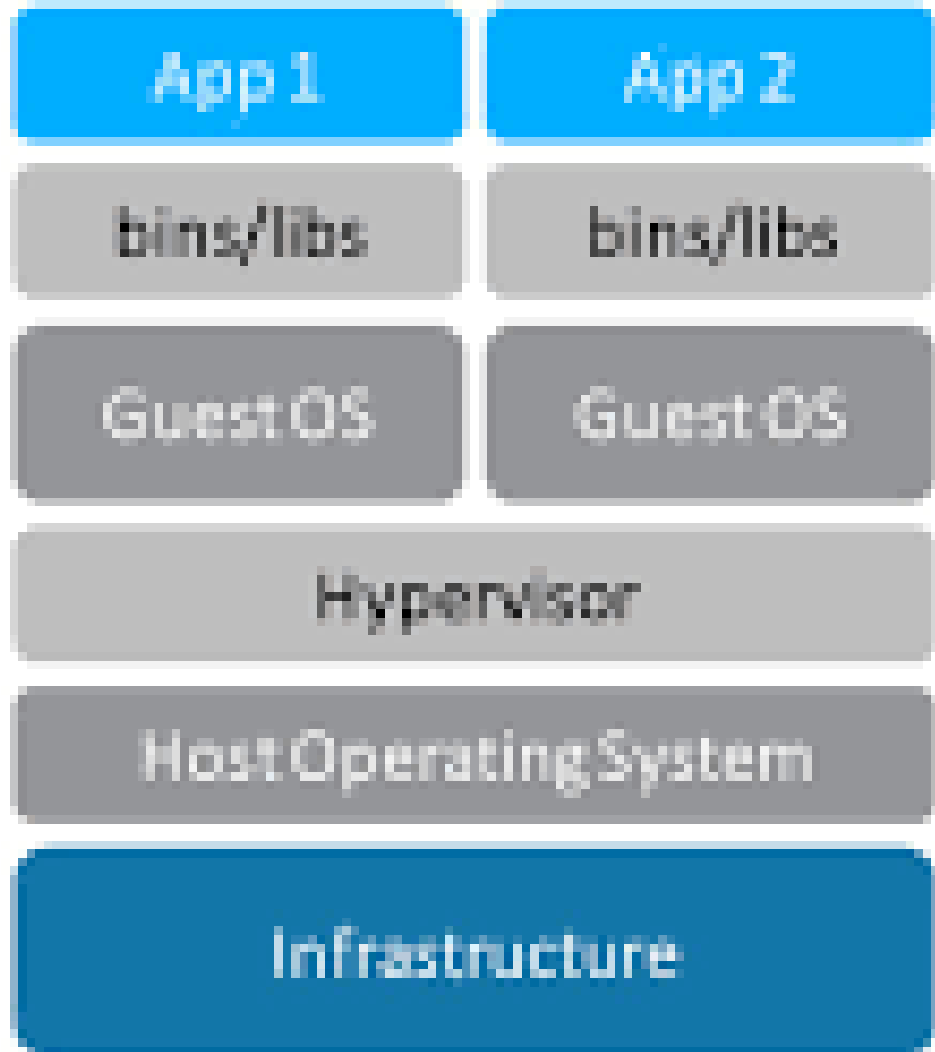
Network



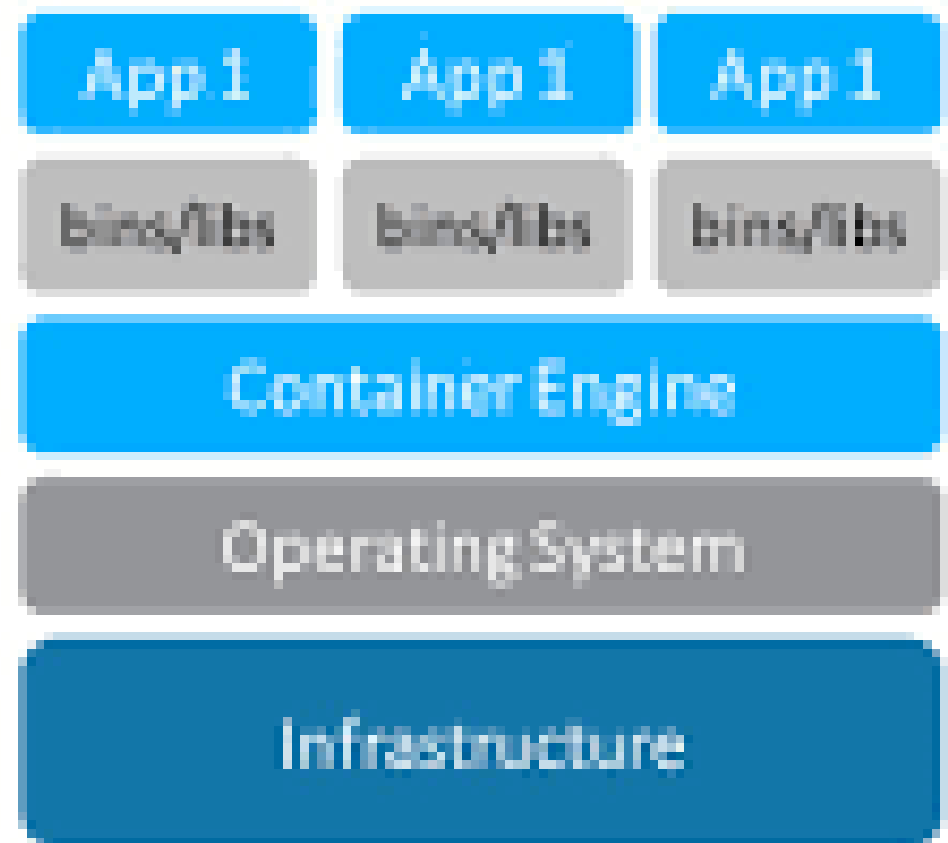
Storage



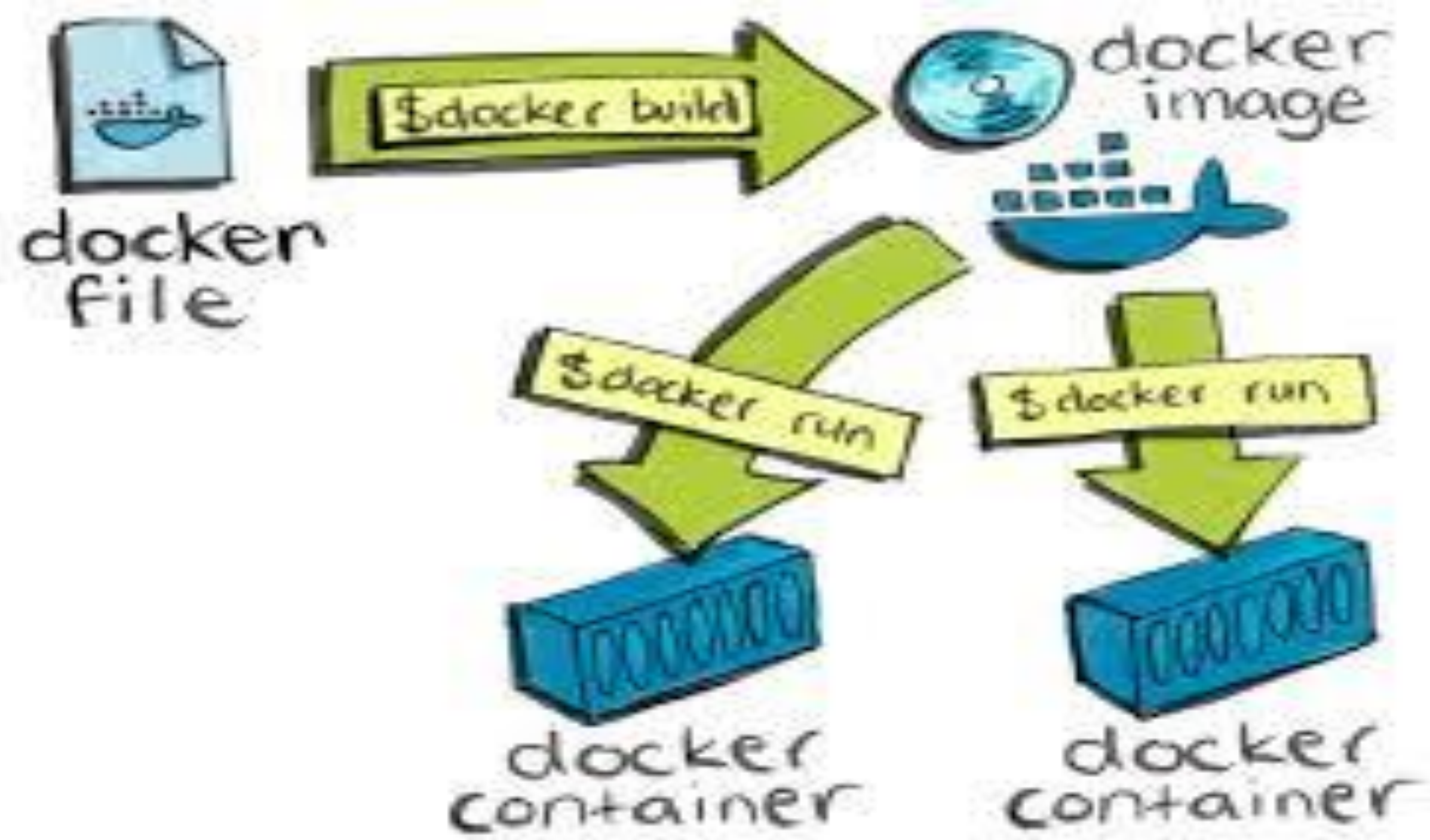
EC2 SOFTLAYER[®]

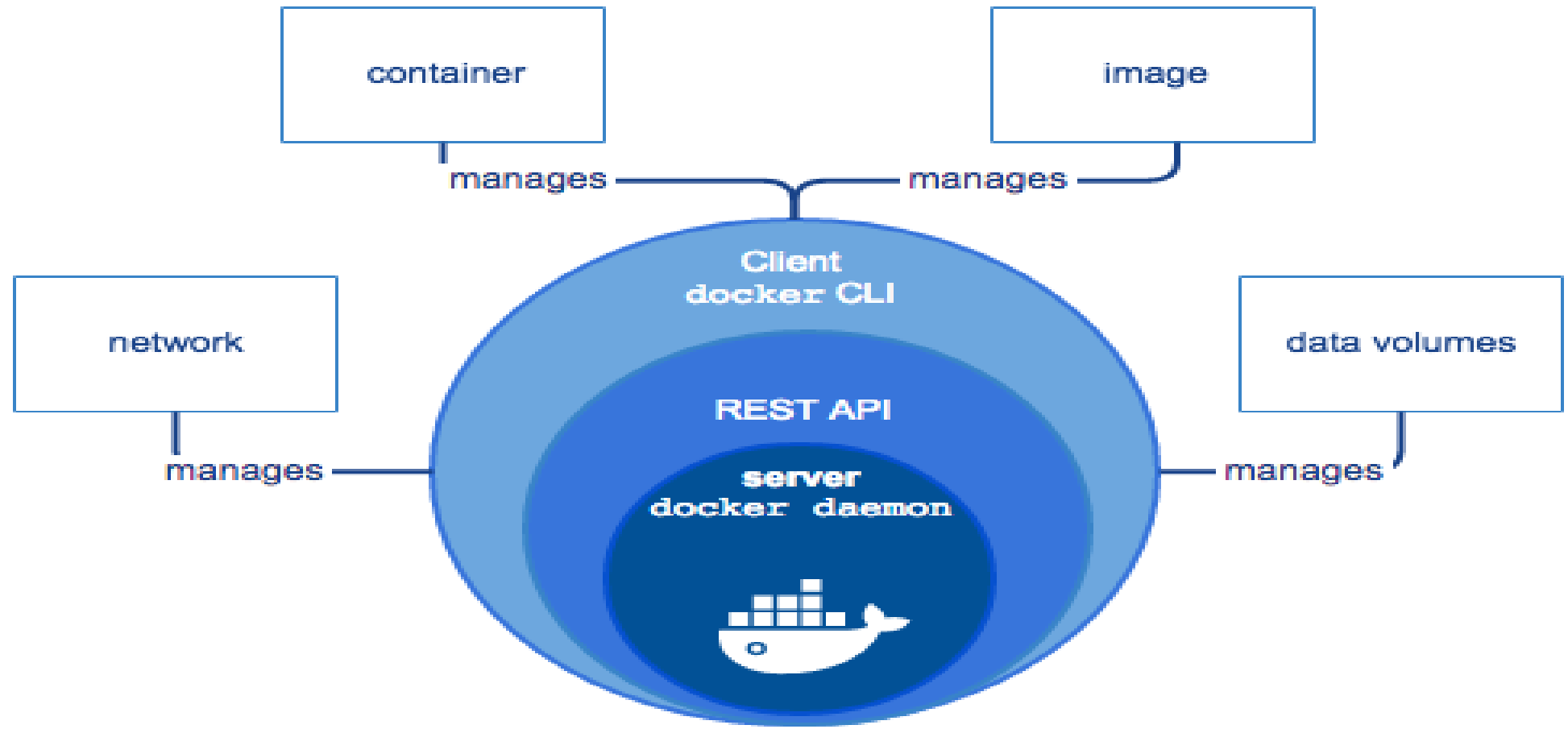


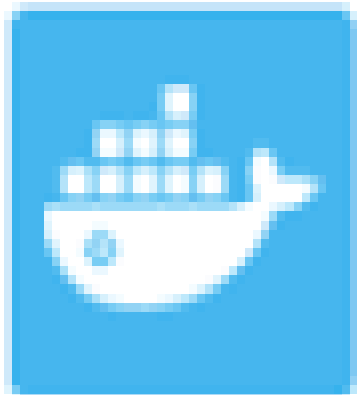
Virtual Machines



Containers



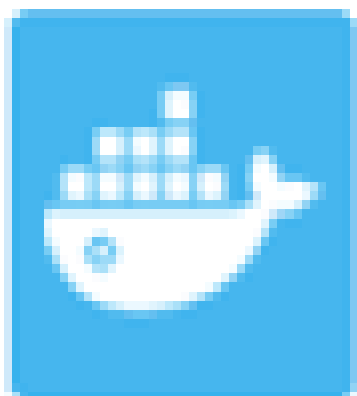




alpine
official

465
STARS

1.1 M
PULLS



ubuntu
official

3.1 K
STARS

39.3 M
PULLS